

Artin Problems 1

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1 Definitions

Problem 1.1. *Show that the image of a representation of dimension 1 of a finite group is a cyclic group*

Proof. Suppose we have a representation of dimension 1 of a finite group G under the field F . $GL_1(F) = F^*$. Since $\forall g \in G, g^{|G|} = e$, thus the homomorphism $f : G \rightarrow F^*$ maps $g \rightarrow f(g)$ where $f(g)^{|G|} = 1$. $\mu_G := \{x \in F \mid x^{|G|} = 1\}$. By claim 1.1a, μ_G is cyclic. $f_*(g) \leq \mu_G$ and all subgroups of a cyclic group are cyclic. Thus $f_*(G)$ is cyclic. $\square$

Claim 1.1a. *Let $G \leq (F^*, *)$ be a finite subgroup. G is cyclic.*

Proof. $|G| = k$. Assume the contrary that G is not a cyclic group. Let $x_1, \dots, x_n$ be generators of G where $m_i = o(x_i)$. Let $\text{lcm}\{m_i \mid i \in \{1, \dots, n\}\} = a$. $(x_1 \dots x_n)^a = (x_1^a \dots x_n^a) = (1 * \dots * 1)$. If $a = k \implies o(x_1 \dots x_n) = k$ and thus G is cyclic. If $a < k$, then there are k solutions to $x^a - 1 = 0$ which has at most a solutions. Thus a contradiction occurs and G is cyclic. $\square$

2 Irreducible Representations

Problem 1.2a. *Choose a suitable basis for $\mathbb{R}^3$ and write the standard representation of the octahedral group O explicitly.*

Proof. The octahedral group is the same as the rotations of a cube. Choose the standard basis of $\mathbb{R}^3$, where e_i intersect a face through their center. If we find the matrices for the generators of $O \cong S_4$ and then the representations of the rest of the elements are going to be products of the representations of the generators. There are 3 generators, rotations of $\frac{\pi}{2}$ along with axes of rotation intersecting the

centers of different faces. So the generating matrices, $\begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos(\frac{\pi}{2}) & -\sin(\frac{\pi}{2}) \\ 0 & \sin(\frac{\pi}{2}) & \cos(\frac{\pi}{2}) \end{bmatrix}, \begin{bmatrix} \cos(\frac{\pi}{2}) & 0 & -\sin(\frac{\pi}{2}) \\ 0 & 1 & 0 \\ \sin(\frac{\pi}{2}) & 0 & \cos(\frac{\pi}{2}) \end{bmatrix}$

and, $\begin{bmatrix} \cos(\frac{\pi}{2}) & -\sin(\frac{\pi}{2}) & 0 \\ \sin(\frac{\pi}{2}) & \cos(\frac{\pi}{2}) & 0 \\ 0 & 0 & 1 \end{bmatrix}$ $\square$

Problem 1.2b. *Choose a suitable basis for $\mathbb{R}^3$ and write the standard representation of the D_n explicitly.*

Proof. Choose the standard basis of $\mathbb{R}^3$, where the n -gon is on the x-y plane and its center of mass is at the origin. If we find the matrices for the generators of D_n and then the representations of the rest of the elements are going to be products of the representations of the generators. Let x and y be generators of D_n . Let $c = \cos(\frac{2\pi}{n})$ and $s = \sin(\frac{2\pi}{n})$.

$$M_x = \begin{bmatrix} c & -s & 0 \\ s & c & 0 \\ 0 & 0 & 1 \end{bmatrix}, M_y = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

□

Problem 2.1. Prove that the standard three-dimensional representation of the tetrahedral group T is irreducible as a complex representation.

Proof. Let $\alpha : G \rightarrow GL_3(\mathbb{C})$

Assume the contrary that $\mathbb{C}^3$ is reducible thus it breaks down into 2 T-invariant subspaces of dimension 1 and 2. Let x, y, z be the basis of $\mathbb{C}^3$ where x is the T-invariant dimension 1 subspace and y and z are in the G-invariant dimension 2 subspace. Since $\text{span}(x)$ is the T-invariant subspace, then if $\forall t \in T, tx = \lambda x$ $\text{span}(x)$ is a g-invariant 1 dimensional subspace. Thus x is a eigenvector of the representations of all the elements of T . Thus $\forall t \in T, t \rightarrow \begin{bmatrix} \lambda_t & & \\ & B_t & \end{bmatrix}$ where λ_t is the eigenvalue cooresponding to x and B_t is the block of the matrix cooresponding to the T-invariant subspace $\text{span}(y, z)$. There exists a map $f : T \rightarrow C^*$ where $t \rightarrow \lambda_t$. $\forall a, b \in T, f(ab) = \lambda_{ab} = \lambda_a \lambda_b = f(a)f(b)$, because the eigenvalue of xy multiply because $\begin{bmatrix} \lambda_a & \\ & B_a \end{bmatrix} \begin{bmatrix} \lambda_b & \\ & B_b \end{bmatrix} = \begin{bmatrix} \lambda_a \lambda_b & \\ & B_a B_b \end{bmatrix} = \alpha(xy)$. Thus f is a homomorphism. However $f_*(T) \subset \mu_{12}$ since $\forall t \in T, t^{|T|} = t^{12} = 1$. Thus f can be simplified from $T \rightarrow C^*$ to $T \rightarrow \mu_{12}$.

Since $T \cong A_4$ and the only normal subgroups of A_4 are the trivial group, K_4 , and A_4 . If the $\ker f = \{e\}$, then f is injective and $A_4 \subseteq \mu_{12}$, and since $|A_4| = |\mu_{12}|$, that would make f an isomorphism and imply A_4 is cyclic which is a contradiction. So $\ker f$ is not e .

If $\ker f = A_4$, then everything gets mapped to 1. So the 1 eigenspaces of the representations of T are all the same because it the eigenvalue of the vector x . The 1 eigenspaces of rotations corresponds to the rotation's axis of rotation. Let u be the vector whose span is the axis of rotation of rotation A through 1 vertex. Let v be the vector cooresponding whose span is the axis of rotation of rotation B through another vertex. Let w be the vector whose span is the axis of rotation of rotation C through a 3rd vertex. u, v, w are in their respective rotations' one eigenspace and are real (given the choice of coordinates). They are linearly independent over the reals due to pointing different directions, thus the

matrix $M = \begin{bmatrix} | & | & | \\ u & v & w \\ | & | & | \end{bmatrix}$ has $\det(M) \neq 0$ in the reals and since each of its entries are real, $\det(M) \neq 0$ over the complex. Thus u, v, w are both linearly independent but also in the span of x which is a contradiction. Thus $\ker f = A_4$ is not a possibility.

Finally if $\ker f = K_4$, then $\forall k \in K_4, f(k) = 1$. The elements in K_4 coorespond to the rotations of π through the opposite edges. Thus the 1 eigenspace, ie the axis of rotations Do the same thing as above, take 3 rotations A, B, C through different vertices and let u, v, w be the vectors whose span

is the axis of rotation. The matrix, $M = \begin{bmatrix} | & | & | \\ u & v & w \\ | & | & | \end{bmatrix}$, has determinant 0 in $\mathbb{R}^3$, because u, v, w point

in different directions. In $\mathbb{C}^3$, since u, v, w have real components, M is a matrix with real entries and thus $\det(M) \neq 0$ in $\mathbb{C}$ as well. Thus u, v, w are linearly independent and also the span of x , meaning they are linearly dependent as well. Thus a contradiction occurs and $\ker f \neq K_4$.

Since $\ker f$ can be none the 3 possibilities, the homomorphism f is not valid. Thus by contradiction, $\mathbb{C}^3$ cannot have a G -invariant subspace of dimension 1 and thus is irreducible. $\square$

Problem 2.2. Consider the standard two-dimensional representation of the dihedral group D_n . For which n is this an irreducible complex representation?

Proof. If $\mathbb{C}^2$ is reducible it must be reducible into 2 1 dimensional subspaces.

Let $c = \cos\left(\frac{2\pi}{n}\right)$ and $s = \sin\left(\frac{2\pi}{n}\right)$. Let x, y be the generators of D_n . Let M_x and M_y be the representations of x, y . $M_x = \begin{bmatrix} c & -s \\ s & c \end{bmatrix}$ and $M_y = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$. $\forall n > 2$, if a vector is G -invariant it must be an eigenvector for all the matrices. Take eigenvectors of M_y , $\begin{bmatrix} 1 \\ 0 \end{bmatrix}$ and $\begin{bmatrix} 0 \\ 1 \end{bmatrix}$. $\forall k \in \mathbb{C}$, $M_x \begin{bmatrix} k \\ 0 \end{bmatrix} = \begin{bmatrix} kc \\ s \end{bmatrix} \notin \text{span}\left(\begin{bmatrix} 1 \\ 0 \end{bmatrix}\right)$ and $M_x \begin{bmatrix} 0 \\ k \end{bmatrix} = \begin{bmatrix} -ks \\ c \end{bmatrix} \notin \text{span}\left(\begin{bmatrix} 0 \\ 1 \end{bmatrix}\right)$. Thus M_x doesn't preserve the 1 or -1 eigenspace of M_y . Thus M_{x^k} doesn't preserve any eigenspace of M_y , thus $\mathbb{C}^2$ has not D_n -invariant subspace of dimension less than 2. For $n = 2$, $D_2 \cong \mu_2$, the x and y axis are preserved because the rotation is a 180 degree rotation thus $\mathbb{C}^2$ is reducible to the x and y axis for D_2 . $\square$

3 Unitary Representations

Problem 3.3a. Let $R : G \rightarrow SL_2(\mathbb{R})$ be a faithful representation of a finite group by real 2×2 matrices with determinant 1. Use the results of Exercise 3.2 to prove that G is a cyclic group.

Proof. Since R is a faithful representations by definition it is injective, thus $G \leq SL_2(\mathbb{R})$. Since $G \leq SL_2(\mathbb{R})$, $\forall g \in G$, $\det(g) = 1$. By 3.2 and as previously proven in bilinear forms, all finite groups of $GL_2(\mathbb{R})$ are conjugate to a subgroup $O(2)$. Thus G is conjugate to $H \leq O(2)$. Thus exists $P \in GL_2(\mathbb{R}) \ni H = PGP^{-1}$. Since conjugation is a isomorphism, $G \cong H$. Furthermore since determinant is a conjugation invariant, thus $\forall h \in H$, $\det(h) = 1$. By Theorem 6.41, from the book the only finite subgroups of $O(2)$ are C_n and D_n . However since $\det(h) = 1$, $\forall h \in H$, $H \leq SO(2)$ and H has no reflection matrices which have determinant -1 and thus cannot be D_n which has reflections matrices and thus must be C_n . $\square$

Problem 3.3b. Determine the finite groups that have a faithful real $2d$ representations

Proof. Let G be a finite group. Let $R : G \rightarrow GL_2(\mathbb{R})$ be a faithful real representation. Thus $G \leq GL_2$ because R is faithful. By 3.2 and as previously proven, all finite groups of $GL_2(\mathbb{R})$ are conjugate to a subgroup $O(2)$. Thus G is conjugate to $H \leq O(2)$. Thus exists $P \in GL_2(\mathbb{R}) \ni H = PGP^{-1}$. Since conjugation is a isomorphism, $G \cong H$. By Theorem 6.41, from the book the only finite subgroups of $O(2)$ are C_n and D_n . Thus G must be either $C_{|G|}$ or $D_{|G|/2}$. $\square$

Problem 3.3c. Determine the finite groups that have faithful real three-dimensional representations with determinant 1.

Proof. Let G be a finite group. Let $R : G \rightarrow SL_3(\mathbb{R})$ be a faithful real representation where representations have determinant 1. Thus $G \leq SL_3(\mathbb{R})$ since R is faithful. Since $G \leq SL_2(\mathbb{R})$, $\forall g \in G$, $\det(g) = 1$. By 3.2 and as previously proven, all finite groups of $GL_3(\mathbb{R})$ are conjugate to a subgroup $O(3)$. Thus G is conjugate to $H \leq O(2)$. Thus exists $P \in GL_2(\mathbb{R}) \ni H = PGP^{-1}$. Since conjugation is an isomorphism, $G \cong H$. Furthermore since determinant is a conjugation invariant, thus $\forall h \in H$, $\det(h) = 1$. Thus $H \leq SO(3)$. By Theorem 6.12.1, the only finite subgroups of $SO(3)$ are $C_k, D_n, T \cong A_4, O \cong S_4, I \cong S_5$ $\square$

Problem 3.4. Let $\langle, \rangle$ be a nondegenerate skew-symmetric form on a vector space V , and let ρ be a representation of a finite group G on V . Prove that the averaging process (10.3.7) produces a G -invariant skew-symmetric form on V , and show by example that the form obtained in this way needn't be nondegenerate.

Proof. G acts on the set of skew symmetric bilinear forms, such that $\langle v, w \rangle_g = \langle gv, gw \rangle$ and $g^* \langle, \rangle = \langle, \rangle_{g^{-1}}$.

$[\cdot, \cdot] := \frac{1}{|G|} \sum_{g \in G} g^* \langle, \rangle$. $\forall h \in G$, $h^* [\cdot, \cdot] = \frac{1}{|G|} \sum_{g \in G} hg^* \langle, \rangle = \frac{1}{|G|} \sum_{g' \in G} g'^* \langle, \rangle = [\cdot, \cdot]$, multiplying by h simply reindexes the elements of G . $\forall v, w \in V$, $[v, w] = \sum_{g \in G} g^* \langle v, w \rangle = \sum_{g \in G} \langle v, w \rangle_{g^{-1}} = \sum_{g \in G} \langle w, v \rangle_{g^{-1}} = - \sum_{g \in G} g^* \langle w, v \rangle = -[w, v]$. Thus $[\cdot, \cdot]$ is skew symmetric. $\forall v \in V$, $[v, v] = -[v, v] \implies [v, v] = 0$

Let $G = \mu_2$. Let $V = \mathbb{R}^2$. Let $A = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$. Let $\langle, \rangle$ be nondegenerate form, $\forall v, w \in \mathbb{R}^2$, $\langle v, w \rangle = v^T A w$.

Let g be the generator of μ_2 , send $g \rightarrow \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ since $g^2 \implies g$ is its own inverse. $[\cdot, \cdot] := \frac{1}{2} \sum_{g \in \mu_2} g^* \langle, \rangle$. $\forall a, b \in \mathbb{R}$, $[e_1, ae_1 + be_2] = a[e_1, e_1] + b[e_1, e_2] = b[e_1, e_2] = \frac{1}{2}[\langle e_1, e_2 \rangle + g\langle e_1, e_2 \rangle] = \frac{1}{2}[1 + \langle e_2, e_1 \rangle] = \frac{1}{2}[1 - 1] = 0$. Thus $[\cdot, \cdot]$ is degenerate. $\square$

Problem 3.5. Problem 3.5 in Artin

Proof. Let $G = C_p = \langle x | x^p = 1 \rangle$ where p is prime. Define a representation which sends $x \rightarrow \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$. $x^n = \begin{bmatrix} 1 & n \\ 0 & 1 \end{bmatrix}$. $\det(x^n - \lambda I) = (\lambda - 1)^2 = 0$, thus the eigenvalues of x^n is 1 for $\forall n \in F_p$. Assume the contrary F_p^2 is reducible, it would break apart into 2 1 dimensional G -invariant subspaces. Since they are 1 dimensional they would be subspaces of the 1 eigenspace.

All elements must preserve the 1 eigenspace of x . Let $\begin{bmatrix} x \\ y \end{bmatrix}$ be the eigenvector of x . $\begin{bmatrix} 1 & n \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} x + ny \\ y \end{bmatrix}$, thus $x + ny \equiv x \implies ny \equiv 0$. $\begin{bmatrix} 1 \\ 0 \end{bmatrix}$ is preserved by all x^n for all $n \in F_p$. However this is the only vector which is preserved because $\forall n \in F_p$, $ny \equiv 0 \iff y = 0$ thus no other vector in the 1 eigenspace of x is preserved by all x^n . Thus there are not 2 G invariant subspaces. Thus there is only 1 G -invariant subspace and thus F_p^2 can not be written as a direct sum of irreducible representations $\square$